



Georgia Institute of Technology

Reflection Deliverable
Bandsaw Monitor

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End of Semester Reflection – Bandsaw Monitor Project

Over the course of this semester, my role in the bandsaw monitor project evolved into a combination of software engineer, system organizer, and technical contributor. I focused heavily on improving the structure and reliability of the system. While the project began as an iteration of an existing safety system, much of my contribution centered around transforming it into a scalable, maintainable, and future-proof solution. I had also planned a redesign and fabrication of new chassis and UI features. This hit a hiccup due to unforeseen circumstances in which I was unable to complete them. I also was working on the project solo so I struggled to maintain the plan without tandem assistance from other team members.

Role and Contribution

My primary role within the team was completing the software refactor and contributing to system-level design improvements. I worked extensively on restructuring the codebase, which initially relied heavily on global variables and hardcoded values. This made debugging and extending functionality extremely difficult. I reorganized the system into modular components which significantly improved readability and maintainability. This then allowed for the integration of Robot unit test cases.

I addressed the absence of testing infrastructure. During a major refactor, there was a high risk of introducing regressions. I helped integrate Robot testing into the workflow, which provided immediate feedback when functionality broke. As noted in the report, this significantly reduced debugging time and increased confidence in making large structural changes. This was a critical contribution because it ensured long-term reliability beyond just this semester.

Overall, my contributions had a direct impact on the project's success by making the system more robust, easier to understand, and more accessible for future developers.

The improved architecture and testing framework transformed the project from a prototype into something closer to a sustainable product.

Skills Development

This project significantly strengthened both my technical and non-technical skills. On the technical side, I developed a much deeper understanding of software architecture, modular design, and testing frameworks. Before this project, I had experience writing functional code, but this was one of the first times I had to think critically about long-term maintainability and scalability.

For example, restructuring the code into distinct modules required me to think about interfaces, data flow, and separation of concerns. Instead of just making code “work,” I had to ensure that each part of the system was logically organized and could evolve independently. Additionally, implementing Robot tests helped me understand the importance of automated testing in real-world systems, especially when working in teams. It also made it much easier for myself when working on the codebase.

On the non-technical side, I improved documentation skills. As I understood the new system structure, I knew it would require clear and effective communication which was then implemented into a cohesive quickstart document.

Team Dynamics

Due to the lack of a team and therefore lack of an experienced prior member, one of the main challenges was the initial lack of documentation and onboarding resources. As mentioned in the report, the project did not have a proper repository or clear setup instructions at the beginning . This created confusion and slowed down progress.

To address this, I contributed to improving documentation and helped establish a more structured workflow. The addition of a GitHub repository and a quickstart guide made a significant difference moving forward. These changes reduced friction and will allow

From this experience, I learned that projects should aim to leave behind clear code, clear fixes, and clear processes. Without these, even an experienced user can struggle.

Individual Growth

This project taught me a lot about myself as both a learner and my ability to adapt and figure things out. I realized that rather than working around issues, we should work to address them at the root, even if it requires more effort upfront. This is because sticking a bandaid over existing problems fails to allow future scalability and flexibility.

A significant area of growth was becoming more comfortable working in ambiguity. At the start of the project, there was a lack of clear direction and documentation. Eventually I found my way through code organization, testing, or documentation allowing me to better understand the project at hand.

Lessons for the Future

If I were to approach a similar project again, I would prioritize setting up proper infrastructure from the beginning. This includes establishing a repository, defining a clear architecture, and implementing testing early on. Doing this upfront would save a significant amount of time and prevent many of the challenges I faced.

I would also push for more consistent user testing throughout development. I really enjoyed the robot tests and thought they really helped to speed up the process for refactoring.

In the future, I will apply these lessons by focusing on scalability and maintainability from the start of any project. Whether in academic, professional, or personal work, I now understand the importance of building systems that are not just functional, but sustainable.

Connection to Course Goals

This project strongly aligns with the course's emphasis on engineering design, iterative development, and teamwork. The transition from an initial prototype to a more refined, modular system reflects the core objective of improving and refining solutions over time.

Additionally, the project reinforced the importance of considering both technical performance and user interaction. The shift from LED indicators to a GUI demonstrates a move toward more user-centered design, which is a key aspect of engineering practice.